

Multiple Choice (10 marks)

1. Most of the oxygen in the atmosphere is made by algae. Where do most of the algae live?
 - (a) coastal ocean
 - (b) deep ocean
 - (c) ponds
 - (d) lakes
2. In the design process, a trade-off occurs when a problem is solved but a feature is sacrificed. Which of the following is an example of a trade-off?
 - (a) A car's gas mileage is increased and the engine has less power.
 - (b) An airplane uses a more efficient engine and has higher performance.
 - (c) A sports drink's taste is improved and has the same nutritional content.
 - (d) A computer company upgrades the hardware and the price remains unchanged.
3. Lithospheric plates have moved to form land formations on the Earth's surface. Which land formation occurs as a result of the plates folding upon one another?
 - (a) an earthquake
 - (b) a mountain range
 - (c) a river
 - (d) a volcano
4. Which changes will most likely have a negative effect on an ecosystem?
 - (a) building a road
 - (b) planting a tree
 - (c) adding a freshwater source
 - (d) creating a nature sanctuary
5. Which geologic process most likely caused the formation of the Mount St. Helens Volcano?
 - (a) converging boundaries
 - (b) diverging boundaries
 - (c) transform faults
 - (d) rift zones

True / False (10 marks)

Answer True or False

6. True or False: Comparing an experiment's results to a control group can help disprove unfounded hypotheses, such as the transformation of hair into worms.
7. True or False: Aphids are first-level consumers that feed on plants in a food web.
8. True or False: Light experiences the same amount of refraction when passing through a telescope lens as it does through a tinted window.
9. True or False: A solution with a pH of 2 can be raised above 7 by adding a base, which neutralizes the acidity.
10. True or False: The sun would be enormous when compared to an average star in size.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the forces acting on a fork as it falls off the edge of a table, explaining how gravity influences its movement toward the floor.
12. Discuss the importance of friction in bicycle ramps and analyze which materials could be used to enhance friction, providing examples and justifications for your choices.

End of Paper